PoxHostAtlas: A reproducible cross-study atlas of host gene-expression remodeling during poxvirus infection

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Abstract

Poxviruses, including vaccinia virus, globally reprogram host gene expression upon infection. Public poxvirus host-response datasets span several viruses, hosts, cell types, and labs. How much of the host response is reproducible across datasets, rather than specific to single experiments, has not been tested systematically. Here we present PoxHostAtlas, a curated collection of public poxvirus host-response RNA-sequencing datasets, reanalysed using a common bioinformatics workflow. Our analysis includes: differential expression, genome-wide DerSimonian-Laird random-effects meta-analysis, leave-one-study-out stability testing of helicase and translation-related genes, host-program over-representation testing, cross-study co-expression network construction to rank host hubs, a leave-one-dataset-out classification stress-test, and independent external validation.

Across four datasets (two vaccinia infection datasets and two Myxoma M003 effector datasets from a single study), 1,834 host genes were reproducibly changed (871 up, 963 down; ≥3 datasets, consistent directionality, meta-FDR < 0.05, $|\log_2 FC| \geq 0.5$). Prominent programs were induction of inflammatory and NF-κB signalling and down-regulation of cell cycle genes; ribosomal protein genes also shifted downward (without over-representation), consistent with host shutoff. In the two vaccinia infection datasets alone, shutoff-related programs stayed down while inflammatory induction was much weaker, suggesting that shutoff is the clearest signal of vaccinia infection itself.

Only 29 of these genes (1.6%) belong to the RNA-helicase, translation-initiation-factor or ribosomal-protein families. DHX15 has the strongest pooled helicase support (meta-FDR 3.6e-19); DHX29 is concordant in direction across all four datasets but heterogeneous (I^2 80%, meta-FDR 0.055), yet outranks DHX15 within vaccinia infection alone. A family-agnostic co-expression network ranks genes outside the curated programs as its top hubs (DTNA, DBP, BTN3A2), with IL6 seventh. A leave-one-dataset-out classifier built on 509 helicase and translation-machinery genes (linear SVM; balanced accuracy 0.79 across three datasets, 0.58 for the held-out vaccinia dataset) performed comparably to housekeeping genes (0.82) and random gene sets (0.74), pointing to broad host remodelling rather than a specific signature. Independent single-cell RNA-seq showed reduced host transcript output in two cell types, and an independent HeLa proteome agreed only partly with the vaccinia-only transcript changes (Spearman $\rho = 0.31$ for 242 transcript-significant genes), although ribosomal and eIF protein levels were not reduced.

PoxHostAtlas is a reusable and falsifiable resource to prioritize host genes for experimental follow-up. As a computational, hypothesis-generating resource, it does not make any causal claims.

Keywords: poxvirus, vaccinia, host shutoff, meta-analysis, reproducibility

Introduction

Vaccinia virus, the prototypic poxvirus and type species of the genus *Orthopoxvirus*, is the best-characterized member of a large family encompassing the causative agents of smallpox and mpox. One of the defining features of poxvirus infection is the induction of host shutoff, in which the virus broadly suppresses host gene expression and protein synthesis while viral mRNAs continue to be translated. Park et al. identified RACK1- and eIF3-centered non-canonical translation initiation during poxvirus-induced shutoff and showed that some host mRNAs escape this shutoff [1]. In addition, host DEAD/DEAH-box RNA helicases such as DHX29 have been shown to regulate the tropism of the oncolytic myxoma virus [2], and DHX29 has been shown to cooperate with eIF3 to facilitate scanning of structured mRNAs [3]. However, there have been few studies looking at the relationship between poxviruses and host RNA helicases.

Although several dedicated poxvirus-host transcriptome studies have been conducted, they can be divided into two groups: (1) larger-scale, virus-centric, long-read sequencing efforts that characterize the transcriptome [4,5]; and (2) a smaller number of studies examining the host response to infection, including bulk RNA-seq together with ribosome profiling [6] and single-cell RNA-seq combined with proteomics [7].

These host datasets come from different studies, using different viruses and cell types, and to our knowledge no integrated analysis has determined which host genes and pathways are reproducibly altered across them; if a gene shows a consistent pattern of change across datasets from different studies, it is less likely to be a batch effect and more likely to represent true biological change. Here, we sought to answer an unbiased, whole-genome question: what host genes and pathways show reproducible alteration across public poxvirus host-response datasets, and what are the most commonly targeted host hubs? We did not limit our search to a particular protein family, because doing so would bias our findings, but instead we focused on the entire genome; we consider host translation and RNA helicase function as a pre-defined subset of the overall question.

Our resulting resource, PoxHostAtlas, combines several public transcriptomic datasets to identify consistently targeted host genes, enabling their prioritization for experimental follow-up; the data were taken directly from publicly available datasets generated by other groups [1,7–10] and re-analyzed here under one common workflow without re-claiming the findings of the original studies, making it a discovery tool rather than a definitive statement on mechanism. Because host responses change as infection progresses, one of our Tier-A datasets includes a time-course experiment (2, 6 and 24 h post-infection), letting us assess how reproducible host alterations develop over the course of infection. Including both vaccinia virus infection and expression of a single Myxoma virus effector (M003) also enables us to ask which host alterations are shared between the two systems and which are specific to one of them.

This reproducibility-first focus was a deliberate methodological decision: rather than highlighting what changed in any given experiment, we focus on what changed consistently across multiple datasets. Risks of

batch effects and other analytical artifacts that could appear as real biological signals in a single dataset are therefore mitigated.

Materials and Methods

Data discovery and stratification

We searched NCBI GEO DataSets ("vaccinia virus RNA-seq", "vaccinia virus ribosome profiling", "myxoma virus RNA-seq" and "poxvirus transcriptome") and PubMed (two queries on DHX29 and one on poxvirus ribosome-profiling and translatome studies) for poxvirus host-transcriptome datasets, retrieving up to 10 records per query; the queries and records are archived in the project repository. Further candidates, including GSE185520 and the long-read vaccinia transcriptome studies [4,5], were identified from the primary literature, article supplementary materials and expert suggestion. All candidates were assessed against predefined selection criteria and tiered by data availability and comparability (Table 1; Table S1). Four datasets met all criteria (Tier A) and were included in the quantitative analyses. Three provided raw count matrices that we re-analysed: GSE278320, vaccinia virus (strain WR) infection of human HAP1 cells from Park et al. [1], and GSE287860 and GSE288000, two GEO sub-series of a single Myxoma virus study [9] in which human cells expressing the Myxoma M003 effector are compared with mCherry-expressing control cells (effector expression, not virus infection). For the fourth, GSE284044, a vaccinia infection time course (2, 6 and 24 h) in Vero cells, an African green monkey kidney cell line [8], we used the authors' deposited DESeq2 result tables. The four Tier-A datasets therefore come from three independent sources and cover one poxvirus infection (vaccinia) and one Myxoma effector. One additional vaccinia dataset (Tier B; GSE185520 [10]) provides only author-processed FPKM and $\log_2$ fold-change tables, so it was not pooled and was used only for directional comparison. Datasets that did not conform to the selection criteria and could not be compared to the other datasets (e.g., viral transcriptome datasets using long reads) were catalogued but not used in further analysis. Categorization prevents the inclusion of inappropriate data in the quantitative model.

For two GSE278320 samples (GSM8544828 and GSM8544830), the GEO sample title indicated a polysome fraction, whereas the supplementary count-file name indicated total RNA. We assumed the file name was correct and recorded the discrepancy. As a sensitivity check, the single-dataset GSE278320 model was refitted without these two samples (5 infected vs. 2 mock) and the translation-factor genes (DHX, EIF, RPL and RPS prefixes) were compared between the two models; all 182 genes significant in either model changed in the same direction. The meta-analysis itself was not recomputed without these samples.

Differential expression and meta-analysis

For the three Tier-A datasets with raw counts, genes with fewer than 10 total reads across the modelled samples were removed, and differential expression was calculated with PyDESeq2 [11], an implementation of the DESeq2 model [12], using a single-factor design (Wald test with Cook's-distance refitting; no fold-change shrinkage). The comparisons were vaccinia- versus mock-infected cells for GSE278320 (the nine

Table 1. Curated poxvirus datasets and their role in the analysis. GEO *n* is the number of samples in the GEO series; the data actually analysed are given in the last column. NTC, non-targeting control.

Dataset	Tier	System / host cells	GEO *n*	Data analysed; role
GSE278320 [1]	A	Vaccinia (WR) infection / human HAP1	38	9 parental HAP1 total-RNA samples (3 mock, 6 infected); meta-analysis, network, classifier
GSE284044 [8]	A	Vaccinia infection / Vero cell line (2, 6, 24 h)	24	Authors' DESeq2 tables; meta-analysis (6 h); time course (2, 6, 24 h)
GSE287860 [9]	A	Myxoma M003 effector vs. mCherry (no infection) / human cells	6	6 samples (3 vs. 3); meta-analysis, network, classifier
GSE288000 [9]	A	Companion sub-series of the same Myxoma M003 study as GSE287860: M003 vs. mCherry in five host backgrounds (NTC and four knockouts)	30	30 samples; meta-analysis (NTC, 3 vs. 3); host-background robustness, network, classifier
GSE185520 [10]	B	Vaccinia infection (wild type and D9/D10 mutant) / human cells	30	Authors' FPKM/$\log_2$FC tables (3 contrasts); directional comparison only
Matía 2024 [7]	External	Vaccinia infection / HeLa and BJ5ta (single-cell); HeLa (proteomics)	—	Supplementary tables: 4,346 HeLa and 11,628 BJ5ta cells; 3 vs. 3 HeLa proteomes; cross-modality validation

parental HAP1 total-RNA samples: 3 mock, 6 infected) and M003 versus mCherry for GSE287860 (3 vs. 3) and for each of the five host backgrounds of GSE288000 (non-targeting control [NTC] and four knockouts; 3 vs. 3 each). Ensembl gene identifiers were collapsed to gene symbols using the annotation supplied with the GSE278320 count files. For GSE284044, the authors' DESeq2 tables (vaccinia vs. mock at 2, 6 and 24 h) were used without re-analysis. No ortholog database was used for these Vero data: their gene symbols were upper-cased and matched to human gene symbols by name. Of the 18,785 Vero genes, 13,028 matched a gene measured in at least one other dataset; the remaining 5,757 (4,510 of them uncharacterized LOC identifiers) did not contribute to pooled estimates.

These comparisons gave 10 harmonized contrasts in total. To avoid pseudoreplication, the meta-analysis used one representative contrast per dataset: GSE278320 (vaccinia vs. mock), GSE284044 (6 h post-infection), GSE287860 (M003 vs. mCherry) and GSE288000 (NTC background). Each gene could therefore be measured in at most four datasets. The other six contrasts (GSE284044 at 2 and 24 h and the four GSE288000 knockout backgrounds) were not pooled; they were used only for the time-course, host-background, forest-plot and heatmap analyses. Duplicate gene symbols within a dataset were combined by inverse-variance weighting. For each gene measured in ≥2 datasets, a DerSimonian-Laird random-effects pooled $\log_2$ fold change [13] and corresponding heterogeneity measures (I^2 [14], τ^2) were calculated along with a directional concordance measure (the fraction of datasets that agree with the majority direction) and a Benjamini-Hochberg adjusted meta-FDR [15]. Genes were considered reproducibly deregulated if they

were measured in ≥3 datasets, changed in the same direction in every dataset that measured them, were significantly deregulated (meta-FDR < 0.05), and had an absolute pooled $\log_2$ fold change ≥ 0.5.

Leave-one-study-out re-pooling and heterogeneity classification were not applied genome-wide. They were run on the 485 genes with DDX, DHX, EIF, RPL or RPS symbol prefixes that were measured in ≥2 datasets (a set that also contains pseudogenes and antisense transcripts; 29 of these genes are among the reproducibly deregulated genes). Each gene was re-pooled after omitting each dataset in turn, and we recorded whether the pooled direction was preserved and whether the re-pooled estimate remained nominally significant ($p < 0.05$). The genes were then assigned to three classes: Class I, all datasets concordant in direction, nominal pooled $p < 0.05$ and $I^2 < 40\%$; Class II, not Class I, ≥75% of datasets concordant, and either nominal $p < 0.05$ or direction preserved in every leave-one-study-out re-pooling (no I^2 criterion); Class III, all remaining genes. Because the classes use nominal rather than FDR-adjusted p-values, class membership does not imply meta-FDR < 0.05.

Enrichment, network-based prioritization, and independent validation

The reproducibly deregulated genes were tested for enrichment in ten host pathways, defined as "programs". Six were explicit gene lists curated from the literature: interferon-stimulated genes, inflammatory/NF-κB signaling, integrated stress response/unfolded protein response, cell cycle/mitosis, cholesterol/sterol biosynthesis and apoptosis. Four were gene families defined by gene-symbol prefix: DEAD/DEAH-box RNA helicases (DDX/DHX), translation initiation factors (EIF), cytoplasmic ribosomal proteins (RPL/RPS) and mitochondrial ribosomal proteins (MRPL/MRPS). Because they are prefix-defined, these families also contain symbols that are not family members, such as ribosomal-protein pseudogenes, antisense transcripts (e.g., DDX11-AS1) and kinases (the RPS6K and EIF2AK genes). Over-representation was tested with a one-sided hypergeometric test against a background of all genes measured in ≥3 datasets (16,841 genes), run separately for all, up-regulated and down-regulated reproducible genes, with Benjamini-Hochberg correction across the ten programs within each set (Table S2); fold enrichment is the ratio of observed to expected overlap, not an effect size. The direction of each program was summarized descriptively as the median pooled $\log_2$ fold change of its genes measured in ≥3 datasets (Table S3). To separate infection-related effects from effector-specific effects, we repeated the random-effects meta-analysis using only the two vaccinia infection contrasts (GSE278320 and GSE284044 at 6 h; genes measured in both) and summarized each program in the same way. With two datasets, this layer cannot meet the ≥3-dataset rule and is reported as a directional sensitivity analysis.

To rank individual host genes without restricting the analysis to any gene family, we built a co-expression network from the 250 reproducibly deregulated genes with the lowest meta-FDR that had an annotated gene symbol (unannotated Ensembl, clone-based and LINC identifiers were excluded). Of these, 228 were present in a 45-sample expression matrix built from the three Tier-A datasets with sample-level counts: GSE278320 (9 samples), GSE287860 (6) and GSE288000 (30, all five host backgrounds). GSE284044 was not included because only the authors' result tables were available. Expression values were $\log_2(\text{CPM}+1)$, computed on the 21,067 genes shared by the three datasets, and each gene was centered on its mean within each dataset.

Pearson correlations were then computed between every pair of the 228 genes across the 45 samples, and two genes were joined by an undirected edge when $|r| \geq 0.7$. For each gene we computed degree centrality on the full network and $|r|$-weighted eigenvector centrality on its largest connected component (NetworkX [16]). Genes were ranked by a hub score equal to the mean of two percentile ranks: connectivity (degree centrality plus eigenvector centrality) and reproducibility ($-\log_{10}$ meta-FDR), so the hub ranking combines network position with meta-analysis significance. Co-regulated modules were identified by greedy modularity maximization [17] on the positive-correlation edges of the largest connected component. Separately, a high-confidence (combined score ≥ 0.7) STRING physical-interaction subnetwork [18] was built only for the 30 top-ranked translation-factor and ribosomal-protein genes of an earlier GSE278320-based prioritization, which was restricted to protein-coding genes using GENCODE v44 biotypes [19] (Figure S10).

Finally, as a portability stress-test, we asked whether host expression could classify samples as infected (or M003-expressing) versus control in a dataset not seen during training. Using the same 45-sample $\log_2$(CPM+1) matrix, before the within-dataset centering (24 infected or M003-expressing, 21 control), four classifiers (elastic-net logistic regression, random forest, gradient boosting and a linear support vector machine [SVM]; scikit-learn [20]) were evaluated by leave-one-dataset-out cross-validation, holding out one whole dataset at a time (three folds, two of which are the Myxoma M003 datasets). Feature sets included all 509 genes with DHX, DDX, EIF, RPL or RPS prefixes and a 29-gene signature (the 30 such genes measured in ≥3 datasets with the lowest meta-FDR, 29 of which were in the matrix; a different set from the prefix-defined genes among the reproducibly deregulated genes). The headline model is the linear SVM on the 509 genes; its 95% bootstrap confidence interval was computed from 2,000 resamples of the pooled out-of-fold predictions. Controls, all with the same linear SVM, were: housekeeping genes as features; interferon-stimulated genes as features; random 509-gene sets (5 draws); within-dataset label permutation (10 permutations); 1,000 random gene sets matched to the 29-gene signature for mean expression and variance; and removal of the 10, 25 or 50 most important genes (ranked by a consensus of elastic-net coefficient magnitude, elastic-net selection frequency and random-forest permutation importance), compared with 1,000 random removals of the same number of genes. To measure batch structure, we also predicted dataset identity with a random forest (3-fold cross-validation accuracy).

The Tier-B dataset GSE185520 was used only for directional comparison: its author-processed $\log_2$ fold changes for three contrasts (wild-type vaccinia at 6 and 18 h, and a D9/D10 decapping-mutant virus at 18 h, each vs. mock) were compared with the pooled estimates by the direction of change of the reproducibly deregulated genes, and by Spearman rank correlation of translation-factor $\log_2$ fold changes (Figure S8; see also Figure S13). We additionally evaluated an independent vaccinia single-cell RNA sequencing and proteomic dataset [7], using the supplementary tables of that study. For the single-cell data (HeLa cells and BJ5ta fibroblasts), we compared the per-cell host-read fraction, number of genes detected and host read counts between infected and uninfected cells (two-sided Mann-Whitney U test). For the proteome (HeLa whole-cell extracts, three infected vs. three uninfected, $\log_2$ LFQ intensities), protein $\log_2$ fold changes were calculated as the difference in mean $\log_2$ intensity and compared with the pooled transcript $\log_2$ fold changes of the vaccinia-only meta-analysis (Spearman correlation and sign agreement), for all matched genes and for genes with vaccinia-only meta-FDR < 0.05. Within the latter set, protein changes of transcript-up and

transcript-down genes were compared with a Mann-Whitney U test. The median protein $\log_2$ fold change was also summarized for each host program across all matched proteins.

Results

A curated public poxvirus atlas

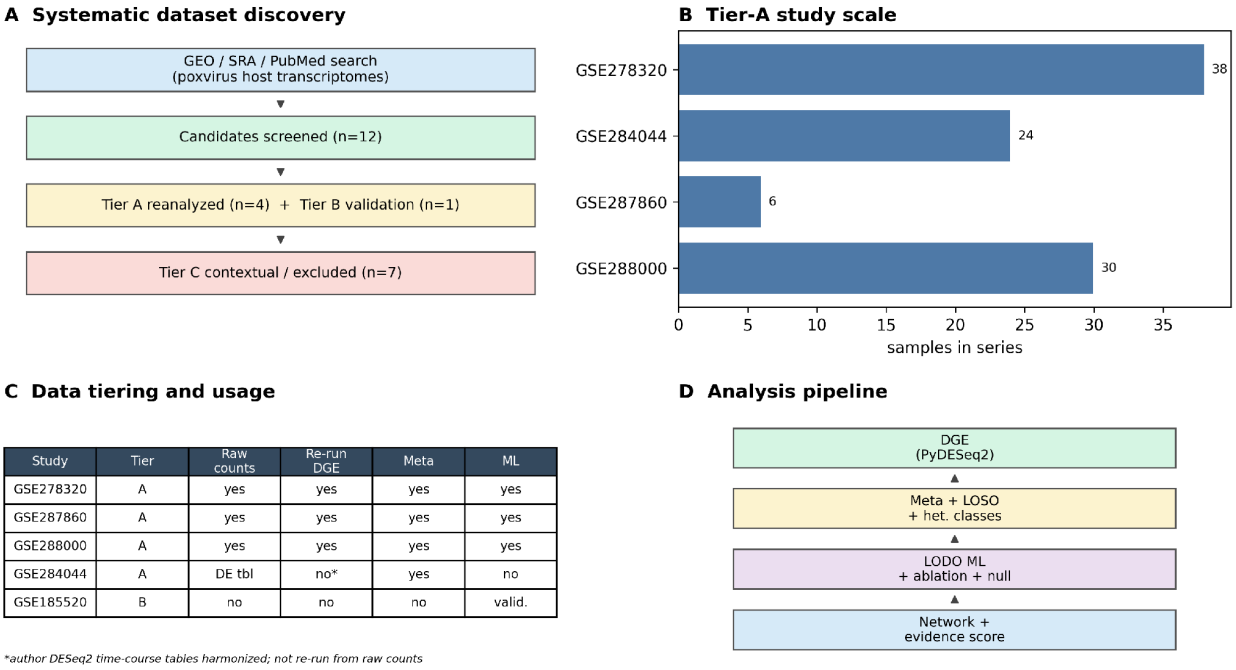
The four Tier-A datasets included in the quantitative analysis are two vaccinia virus infection datasets (human HAP1 cells [1]; a 2, 6 and 24 h time course in the African green monkey Vero cell line [8]) and two GEO sub-series of one Myxoma virus study that compare human cells expressing the M003 effector with mCherry controls [9]. The Tier-B dataset is one further vaccinia study, used only for directional comparison [10] (refer to Table 1 and Figure 1). These Tier-A experiments yielded 10 harmonized contrasts; one per dataset entered the meta-analysis (GSE284044 at 6 h; GSE288000 in the non-targeting-control background), and the rest were used for temporal and host-background comparisons. In total, 16,841 host genes were measured in at least three datasets; any three of the four datasets include at least one vaccinia and one M003 dataset, and therefore data from at least two laboratories.

A broad host program is reproducibly remodeled across poxvirus datasets

A total of 1,834 host genes were reproducibly remodeled (measured in ≥3 datasets, same direction in each of them, meta-FDR < 0.05, |pooled $\log_2$FC| ≥ 0.5): 871 were up-regulated and 963 down-regulated, a subset of the 3,337 genes with meta-FDR < 0.05 in Figure 2. Only 29 of them (1.6%) were DEAD/DEAH-box RNA helicases, translation initiation factors or ribosomal-protein genes (8, 7 and 14, the last including two pseudogenes), so no single gene family dominated the signal. Rather, two host programs stood out among the reproducible genes (Figure 3). Inflammatory and NF-κB signaling genes (e.g., IL6, IL1B, CXCL8, and BIRC3) were over-represented 5.4-fold (13 of 22; false discovery rate [FDR] = 3e-7), all of them up-regulated. Cell-cycle and mitotic genes (e.g., CDK1, CCNB1/CCNB2, PLK1, and FOXM1) were over-represented 5.7-fold (13 of 21; FDR = 3e-7), all of them down-regulated (median pooled $\log_2$FC across the 21 measured genes, −0.80). Ribosomal-protein genes also tended to decrease: 158 of the 189 measured RPL/RPS-prefixed genes had a negative pooled $\log_2$FC, and 24 of the 26 with meta-FDR < 0.05 were down-regulated. This prefix-defined family, which also contains pseudogenes and RPS6K kinases, was not itself over-represented among the reproducible genes (14 of 189; 0.68-fold; Table S2). Among the most strongly induced genes were inflammatory transcripts (IL6, IL1B, and BIRC3), while the most strongly repressed were proliferation- and metabolism-related (e.g., IGF1 and DEPTOR; Table S5).

Because two of the four datasets are Myxoma M003 effector contrasts from a study of an effector-triggered NF-κB response [9], we recalculated the program directions using only the two vaccinia-infection datasets (GSE278320 and GSE284044 at 6 h; Table S3). The clearest signature in this case was host shutoff (with its ribosomal proteins median $\log_2$ fold-change close to −0.79, cell cycle about −0.58, and translation initiation factors equal to about −0.32), while the inflammatory induction, although present, was considerably lower

Figure 1. PoxHostAtlas: systematic public poxvirus host-response atlas, tiering, and pipeline



Study	Tier	Raw counts	Re-run DGE	Meta	ML
GSE278320	A	yes	yes	yes	yes
GSE287860	A	yes	yes	yes	yes
GSE288000	A	yes	yes	yes	yes
GSE284044	A	DE tbl	no*	yes	no
GSE185520	B	no	no	no	valid.

**author DESeq2 time-course tables harmonized; not re-run from raw counts*

Figure 1. Systematic dataset discovery, Tier-A dataset scale, data tiering and use, and the end-to-end analysis pipeline.

(about +0.74 vs. +2.0 in the aggregate). With only two datasets, few program genes reach FDR < 0.05 individually (e.g., 2 of 84 ribosomal-protein genes), so this is a directional sensitivity analysis. Therefore, within vaccinia infection the most consistent signature is host shutoff; the stronger inflammatory induction seen in the aggregate is driven partly by the Myxoma effector contrasts.

As one Tier-A dataset is a time-course study of vaccinia (2, 6, and 24 h post-infection), it was possible to follow the shutoff signature over time rather than at a single time point. Cell-cycle genes were directionally lower than in mock-infected cells at all three time points (median $\log_2$FC −0.40, −0.32 and −1.61), whereas ribosomal-protein genes were unchanged at 2 h (+0.03), leaned down at 6 h (−0.28) and were clearly down at 24 h (−0.74). Significant decreases in both programs were confined to 24 h (18 of 21 cell-cycle and 7 of 84 ribosomal-protein genes at FDR < 0.05). The shutoff signature therefore builds over the course of infection, pointing toward a progressive rather than transient host response. This result should be interpreted with caution, because it is based solely on one study in a monkey cell line whose gene symbols were matched to human symbols by name, but the direction of both programs agrees with the human datasets.

Network ranking and the helicase sub-program

A family-agnostic co-expression network (228 nodes and 4,430 edges; Figure 4) contained a co-up-regulated module of 128 genes and a co-down-regulated module of 93 genes. A hub score combining network connectivity with reproducibility ranked DTNA, DBP and BTN3A2 highest, and 11 of the 12 top-ranked hubs lie outside the curated host programs (Table 2). IL6 was the highest-ranked curated-program gene (7th of 228),

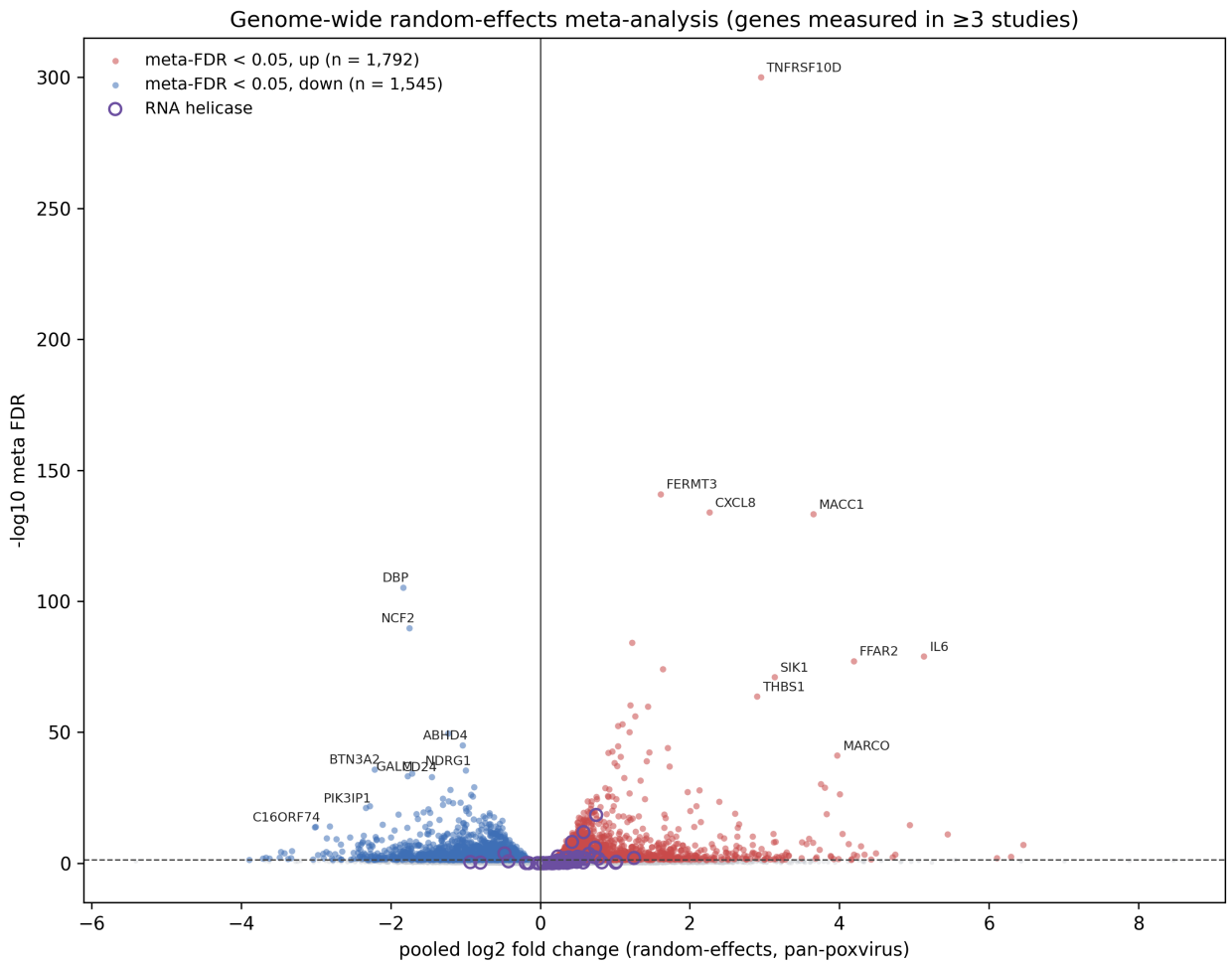


Figure 2. Genome-wide random-effects meta-analysis volcano for the 16,841 host genes measured in ≥3 datasets. Coloured points have meta-FDR < 0.05 (1,792 up, 1,545 down); the 1,834 reproducibly remodeled genes are the subset that also has the same direction in every dataset that measured them and $|\text{pooled } \log_2\text{FC}| \geq 0.5$, and labelled genes are drawn from this subset. RNA helicases (DDX/DHX) are circled, not pre-selected.

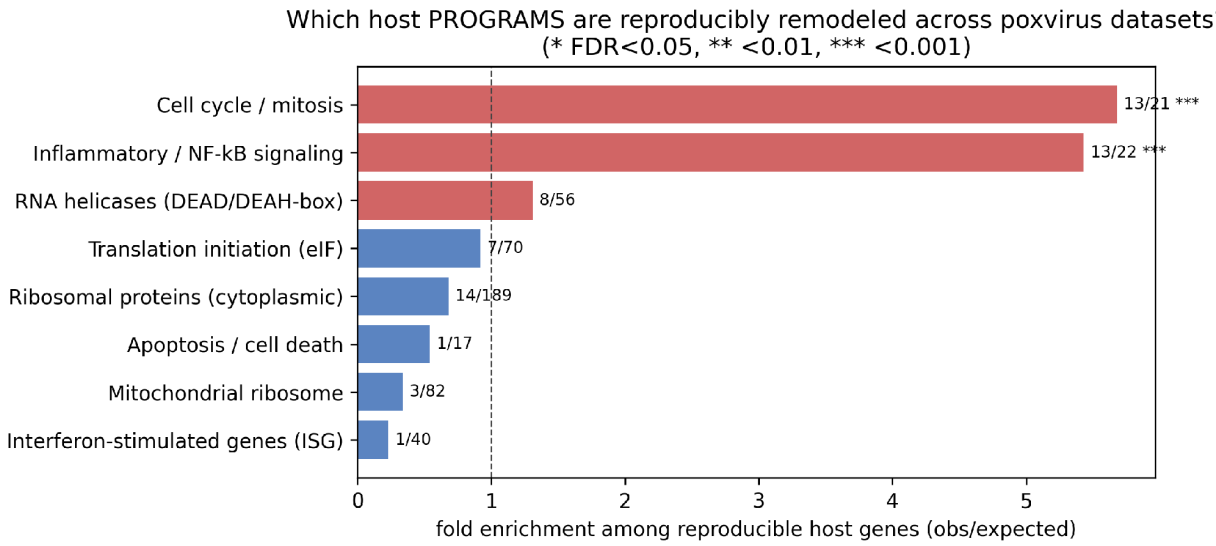


Figure 3. Fold over-representation (observed/expected) of host programs among the 1,834 reproducibly remodeled genes, tested against all genes measured in ≥3 datasets (labels, hits/program size; asterisks, Benjamini-Hochberg FDR; programs without hits not shown). Bar colour marks over- (red) or under-representation (blue), not direction: inflammatory/NF-κB hits are all up-regulated and cell-cycle/mitosis hits all down-regulated (Table S2). Prefix-defined programs (DDX/DHX, EIF, RPL/RPS, MRPL/MRPS) can include pseudogenes and other non-member genes.

and MAFB and PTPRC ranked 14th and 18th (Figure S9). Of the DDX/DHX helicase genes with meta-FDR < 0.05, 17 of 18 were up-regulated, the exception being DHX58; a 19th gene matching the family prefix, the antisense transcript DDX11-AS1, was also up-regulated but is not a helicase. The family was not over-represented among all reproducible genes (1.3-fold; FDR 0.87) but was enriched among the up-regulated ones (2.8-fold; FDR 0.039; Table S2). Within the family, the lowest meta-FDR was that of DHX15 (3.6e-19; top 0.6% of genes measured in ≥3 datasets), which was up-regulated in all four datasets (pooled $\log_2$FC 0.75), with about two-thirds of its random-effects weight coming from the two Myxoma M003 datasets. DHX29 was also up-regulated in all four datasets but was highly heterogeneous (I^2 about 80%) and did not reach significance after correction (meta-FDR approximately 0.06); like DHX15, it falls in the conserved, high-heterogeneity class rather than the context-dependent class (Class II in Table 3; both genes are marked in Figure 5). When only the two vaccinia infection datasets were pooled, the order reversed: DHX29 was significant (pooled $\log_2$FC 0.83, meta-FDR 0.006, I^2 23%) while DHX15 was not (meta-FDR 0.075). DHX15 therefore has the stronger pooled support and DHX29 the stronger support within vaccinia infection; both are candidates for follow-up.

Reproducibility across studies, modalities and cell types

As a portability stress-test, we used leave-one-dataset-out classification to ask whether infected (or M003-expressing) samples could be told apart from controls in a dataset the model had not seen. This was possible only for the three Tier-A datasets with sample-level counts (45 samples: GSE278320, 9; GSE287860, 6; GSE288000, 30), so GSE284044 was not included. A linear support vector machine trained on the 509

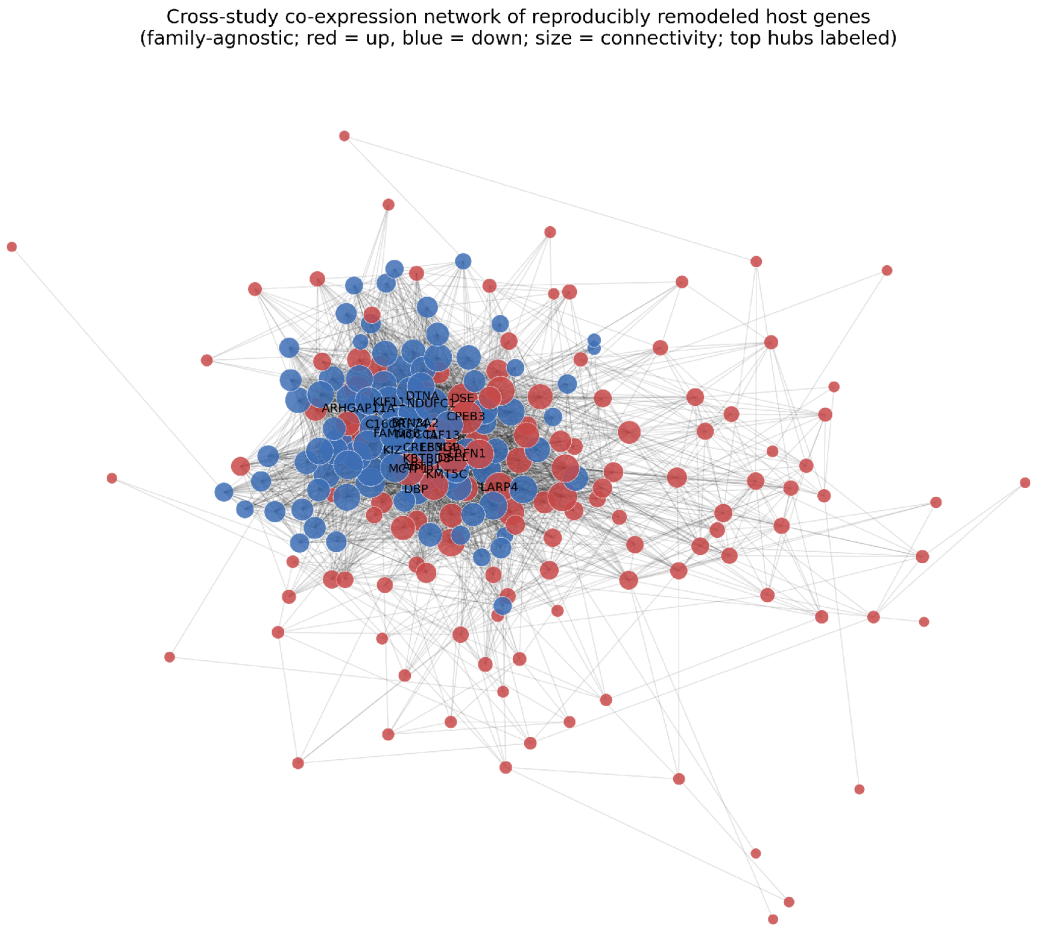


Figure 4. Cross-study co-expression network of reproducibly remodeled host genes. The largest connected component is shown (223 of the 228 network genes; the other five have no edges). Edges join gene pairs with $|r| \geq 0.7$ (Pearson) across 45 samples from GSE278320, GSE287860 and GSE288000, centered within each dataset (GSE284044 is not included: it entered the analysis as author-deposited DESeq2 result tables rather than a sample-level count matrix). Node color gives the direction of change (red, up; blue, down) and node size increases with degree. Labels mark the 22 genes with the highest degree; this is not the same as the hub-score ranking, which is given in Table 2 and Figure S9.

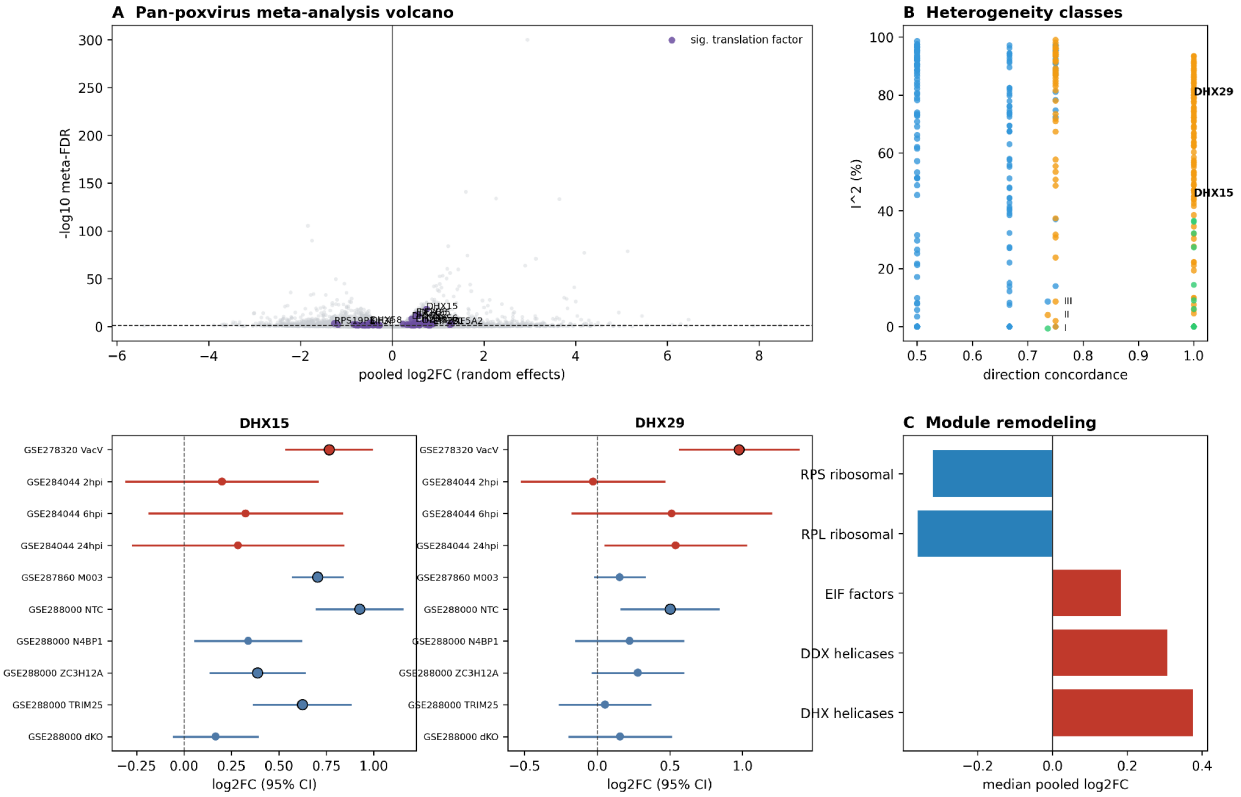


Figure 5. Translation/helicase sub-program. (A) Meta-analysis volcano across the four datasets (genes measured in ≥3 datasets) with significant translation-related genes (DDX/DHX, EIF, RPL/RPS; meta-FDR < 0.05) highlighted. (B) Heterogeneity classes (I^2 versus direction concordance; green, Class I; orange, Class II; blue, Class III) for the 485 genes of these families measured in ≥2 datasets, with DHX15 and DHX29 marked. Bottom left and center: $\log_2$FC (95% CI) for DHX15 and DHX29 in each of the 10 harmonized contrasts (red, vaccinia; blue, Myxoma M003; outlined points, padj < 0.05). Both genes are up-regulated in all four meta-analysis datasets and fall in Class II; DHX15 is significant after pooling, whereas DHX29 is highly heterogeneous and not significant. (C) Median pooled $\log_2$FC by module (RPS, RPL, EIF, DDX, DHX).

Table 2. Top-ranked reproducible host hubs prioritized for experimental follow-up.

Gene	Program	Direction	Pooled $\log_2$FC	meta-FDR	Degree	Hub score
DTNA	other	up	+1.64	8.4e-75	89	0.956
DBP	other	down	−1.84	5.4e-106	82	0.952
BTN3A2	other	down	−2.22	1.4e-36	99	0.941
LARP4	other	up	+1.04	2.2e-45	82	0.921
DSE	other	up	+1.34	2.2e-32	84	0.904
ABHD4	other	down	−1.04	8.0e-46	74	0.893
IL6	Inflammatory/NF-kB	up	+5.13	1.2e-79	68	0.884
GALM	other	down	−1.77	5.4e-34	73	0.875
SATB1	other	down	−0.90	2.8e-26	79	0.871
MACC1	other	up	+3.65	6.1e-134	62	0.866
ABTB1	other	down	−1.50	1.9e-21	96	0.864
LENG9	other	up	+2.00	1.1e-20	95	0.860

Genes are ranked by hub score among the 228 co-expression network genes (of the 250 named reproducible genes with the lowest meta-FDR, the 228 quantified in the 45-sample expression matrix), not among all 1,834 reproducible genes. Hub score = mean of the percentile rank of connectivity (degree centrality + eigenvector centrality) and the percentile rank of reproducibility ($-\log_{10}$ meta-FDR); ties are ordered by degree (KMT5C, 13th, has the same hub score as LENG9). Degree = number of co-expression edges with $|r| \geq 0.7$ (either sign) out of 227 possible partners; the highest degree in the network is 112. 'other' = not in the curated inflammatory/NF-κB, interferon-stimulated, cell-cycle/mitosis, RNA-helicase, eIF or ribosomal-protein gene sets. The full ranking is available in the project repository.

translation-family genes (DDX/DHX, EIF, RPL/RPS symbols) present in this matrix reached a mean balanced accuracy of 0.79 across the three folds (95% bootstrap CI 0.67–0.90 on the pooled out-of-fold predictions), but this was carried by the two Myxoma M003 datasets (1.00 and 0.80); on the held-out vaccinia dataset it was 0.58, close to chance, and the other three algorithms had lower mean accuracies over the three folds (0.44–0.71). With labels permuted within each study, performance fell to chance (0.54, mean of 10 permutations), and the identity of the study was itself highly predictable (3-fold cross-validated accuracy 0.93, against 0.67 for always guessing the largest study), confirming a strong batch structure. Removing the 25 top-ranked features degraded the model more than 99.8% of random 25-gene removals ($p = 0.003$). However, 15 housekeeping genes (0.82) and random 509-gene sets (0.74, mean of five sets) classified infection about as well, and a 29-gene signature of the most significant translation-family genes in the pooled meta-analysis — a different set from the 29 such genes among the 1,834 reproducible genes, with 14 in common — did not outperform 1,000 expression- and variance-matched random gene sets (0.79; 81.7th percentile; $p = 0.18$; Table S6). Infection status therefore transferred between the two Myxoma datasets but not to the held-out vaccinia dataset, and it did not depend specifically on this gene set, which is what a broad host program would look like.

Leave-one-study-out re-pooling and the heterogeneity classification were run on a smaller, pre-defined gene set: the 485 translation-family genes (DDX/DHX, EIF, RPL/RPS symbols, including pseudogenes) measured in at least two Tier-A datasets (the same symbol classes as the 509-gene classifier set above, counted in the meta-analysis rather than in the 45-sample matrix). Only 29 of these are among the 1,834 reproducible genes, so these results describe the translation machinery, not the genome-wide program. Of the 485 genes, 33 met the strict Class I rule (same direction in every dataset, nominal $p < 0.05$ and $I^2 < 40\%$), 254 fell in the broader Class II (same direction in at least 75% of datasets and either nominally significant or direction-stable when any one dataset was removed) and the remaining 198 in Class III, which keeps the same direction in at least half the datasets but meets neither of the stricter rules (Table 3, Figure 6B). In leave-one-study-out re-pooling, 327 of the 485 genes (67.4%) kept their effect direction every time, but only 28 (5.8%) stayed nominally significant every time (Table S4); DHX15 retained direction and significance in all four re-poolings.

Genome-wide, rank concordance between the datasets was modest (Kendall's W = 0.40 over the 10,976 genes measured in all four; Figure 6A). The two Myxoma M003 datasets, which are sub-series of the same study, were highly concordant (Spearman $\rho = 0.87$), but every correlation between different sources was weak: $\rho = 0.21$ between the two vaccinia datasets, 0.17 and 0.16 between GSE284044 and the two M003 datasets, and −0.08 and −0.13 between GSE278320 and them. Most of the overall rank agreement therefore comes from two sub-series of one study. In the Tier-B dataset GSE185520 [10], for which only author-processed FPKM-based fold changes were available, support was mixed: of the roughly 1,370 reproducible genes measured there, 61%, 55% and 64% changed in the same direction in its three contrasts (wild-type vaccinia at 6 and 18 h and a D9/D10 mutant virus at 18 h), and at 18 h in wild-type infection the fold changes were globally negative (91% of all measured genes lower; median $\log_2$FC −2.08 over all genes). DHX15 was not induced in this dataset ($\log_2$FC −0.01 and −2.77 for wild-type virus at 6 and 18 h, +0.10 for the mutant; Figure S13).

Finally, we asked whether host shutoff is also seen in other cell types and at the protein level, using an independent vaccinia single-cell RNA-seq and proteomic dataset [7]. In single cells, the median fraction of reads from host genes fell from 0.998 in uninfected to 0.758 in infected HeLa cells (2,172 vs 2,174 cells) and from 0.999 to 0.804 in BJ5ta fibroblasts (6,826 vs 4,802 cells), and fewer genes were detected per cell (median 4,978 vs 1,976 in HeLa; about 2,340 vs 1,740 in BJ5ta; Mann–Whitney $p < 10^{-300}$ for each comparison; Figure S14). In the HeLa whole-cell proteome (three infected vs three uninfected samples), protein and transcript changes were positively correlated for the 242 genes that were significant in the vaccinia-only meta-analysis and also detected in the proteome (Spearman $\rho = 0.31$, p = 8.1e-7; same direction for 64%), and genes that were down at the transcript level had lower protein levels than genes that were up (median protein $\log_2$FC −0.21 vs +0.05, p = 1.6e-5; Figure 7, reproduced as Figure S15). Across all 7,111 matched genes the correlation was weak ($\rho = 0.08$) and the median protein $\log_2$FC was −0.08. Against that background, cell-cycle proteins were modestly lower (median −0.17, n = 13), whereas ribosomal-protein (+0.03, n = 79) and eIF (+0.14, n = 44) protein levels were not reduced. These data support reduced host transcript output in two cell types and a modest transcript–protein agreement, but they do not show a loss of translation-machinery proteins.

Table 3. Reproducibility classification and leave-one-study-out stability of the 485 translation-family genes (DDX/DHX, EIF, RPL/RPS symbols, including pseudogenes) measured in at least two Tier-A datasets.

Class	Definition	Genes
I — conserved, low heterogeneity	Same direction in every dataset, nominal pooled $p < 0.05$ and $I^2 < 40\%$	33
II — conserved, high heterogeneity	Same direction in ≥75% of datasets and either nominal $p < 0.05$ or direction kept in every leave-one-study-out re-pooling (no I^2 condition)	254
III — context-dependent	All remaining genes (same direction in ≥50% of datasets)	198
Total classified		485
Direction preserved in every leave-one-study-out re-pooling		327 (67.4%)
Nominal significance preserved in every re-pooling		28 (5.8%)
DHX15	Direction and significance preserved in all four re-poolings	✓

Class names follow the analysis code. Class II has no I^2 condition: 95 of its 254 genes have $I^2 < 40\%$. No gene fell in Class IV, which requires the direction to agree in fewer than half the datasets. Only 29 of the 485 genes are among the 1,834 reproducible genes.

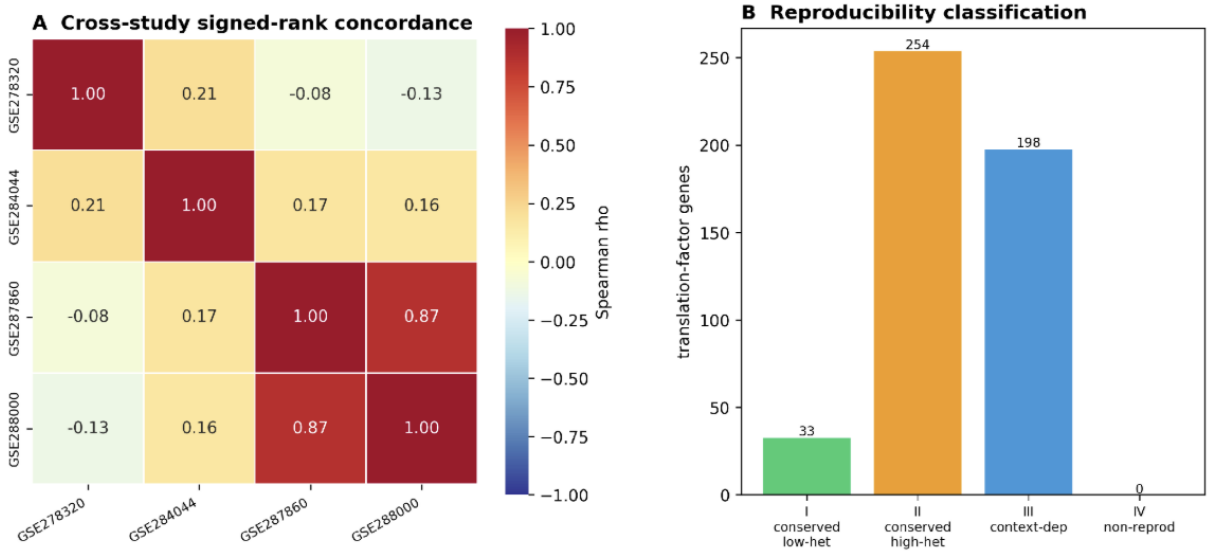


Figure 6. Cross-study rank concordance and reproducibility classes. (A) Spearman correlation between the four Tier-A datasets of signed $-\log_{10}$ p-values over the 10,976 genes measured in all four (Kendall's W = 0.40). (B) Reproducibility classes of the 485 translation-family genes (DDX/DHX, EIF, RPL/RPS) measured in at least two datasets (definitions in Table 3); no gene fell in Class IV, so that bar is empty.

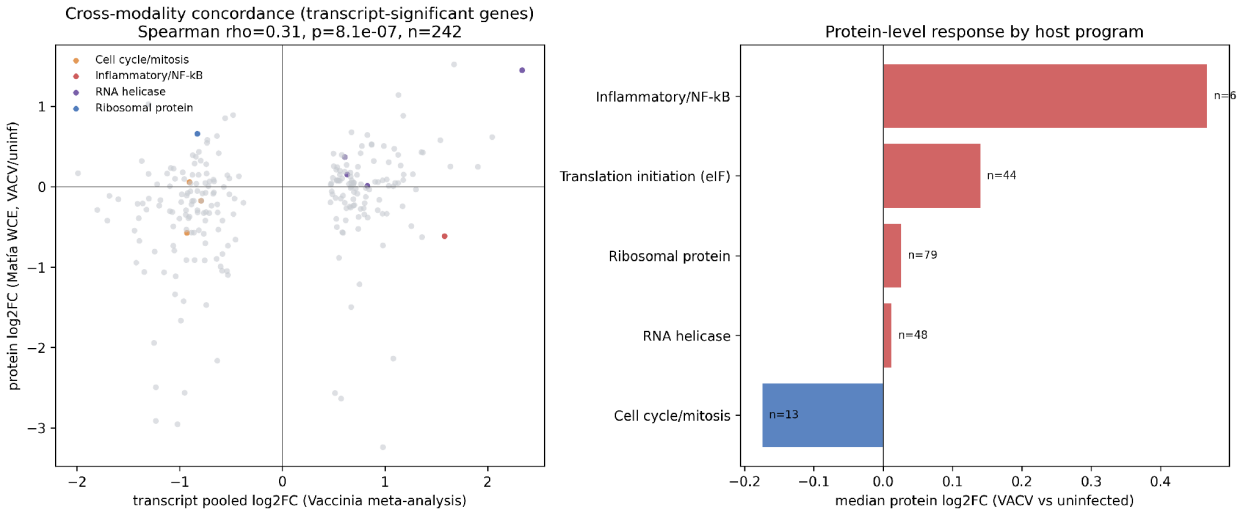


Figure 7. Independent external validation in the Matía et al. vaccinia proteome [7]. Left: HeLa protein $\log_2$ fold change (infected vs uninfected) against pooled transcript $\log_2$ fold change for the 242 genes that were significant in the vaccinia-only meta-analysis and also detected in the proteome (Spearman ρ = 0.31, p = 8.1e-7). Right: median protein $\log_2$ fold change of all matched proteins in each host program; ribosomal-protein and eIF protein levels were not reduced.

When all the evidence is taken together, no single dataset determined the direction of these changes: by the reproducibility rule, each of the 1,834 reproducible genes changed in the same direction in every dataset that measured it, and 67.4% of the translation-family genes kept their pooled direction when any one dataset was removed. Statistical significance, however, often depended on individual datasets (only 5.8% of these genes stayed significant in every re-pooling), and effect sizes varied between datasets (median I^2 55% across the 1,834 genes). Inflammatory genes such as IL6 and IL1B were among the most strongly induced but also among the most heterogeneous (I^2 79% and 96%), whereas cell-cycle genes — the most consistently repressed group, 13 of the 21 tested and 10.8-fold over-represented among the down-regulated reproducible genes — were less heterogeneous (median I^2 41%).

Discussion

Poxvirus infection causes host shutoff, in which host protein synthesis is broadly inhibited while viral mRNAs and some host mRNAs continue to be translated [1]. Across the datasets analyzed here, the host response also included repression of cell-cycle genes and induction of inflammatory and NF-κB genes. Cell-cycle repression recurs, at least directionally, in all four datasets, which come from three independent studies, so it is unlikely to reflect the conditions of one laboratory. The inflammatory signal is driven partly by the Myxoma M003 effector contrasts (median pooled $\log_2$FC of these genes +2.0 over the four datasets, but +0.74 in the vaccinia-only pooling, where fewer of them are measured), consistent with the source study's focus on an effector-triggered NF-κB response [9]. Among the helicases, DHX15 has the strongest pooled support, largely from the M003 datasets, whereas DHX29 is better supported within vaccinia infection; both warrant further research.

In biological terms, the reproducible program makes sense. Repression of cell-cycle genes, together with a directional decrease of ribosomal-protein genes (of the 189 measured, 26 reached meta-FDR < 0.05 and 24 of those were down), fits a virus that takes over the host's translation machinery while causing the host cells to stop dividing so that the virus can replicate. Translation-initiation factor genes decreased only in the vaccinia-only analysis, and only directionally (median $\log_2$FC about -0.32). Unlike inflammatory genes, interferon-stimulated genes were not reproducibly induced (1 of 40 met the criteria; 0.23-fold of expectation). Cell-cycle repression and the ribosomal-protein decrease appear, at least directionally, in vaccinia infection of human and monkey cells, and cell-cycle repression and inflammatory induction also appear in the M003 effector contrasts, so they are candidate common features of poxvirus–host interaction. Whether they extend to other orthopoxviruses, such as monkeypox virus, which causes mpox and is a growing public health threat, remains to be tested.

Limitations

All primary signals are at the transcript level; the only protein-level evidence is one independent proteomic dataset [7], in which ribosomal-protein and translation-initiation proteins were not reduced (median $\log_2$FC +0.03 and +0.14). Ribosome profiling and perturbation experiments are needed for any mechanistic claim. Only two of the four Tier-A datasets are vaccinia infections: GSE287860 and GSE288000 are two GEO sub-series of one Myxoma study [9] comparing cells that express the M003 effector with mCherry controls, which is effector expression, not infection. Because they were analyzed as separate datasets, the $\geq$3-dataset criterion can be met by one vaccinia dataset plus these two sub-series, which is the case for 592 of the 1,834 reproducible genes. We therefore also report the vaccinia-only layer, which, with only two datasets, cannot enforce that criterion and is reported as a directional sensitivity analysis.

For GSE284044 we used the authors' deposited DESeq2 tables rather than re-analyzing counts, and its Vero gene symbols were matched to human symbols by name without an ortholog database, so roughly 30% of its genes, mostly uncharacterized LOC identifiers, matched no gene in the human datasets; public metadata may also contain errors. Leave-one-study-out re-pooling and the Class I–III classification covered only the 485 DDX/DHX/EIF/RPL/RPS-prefixed genes measured in $\geq$2 datasets, and re-pooling never removes both M003 sub-series at once. The classifier separated infected from control samples mainly in the two Myxoma datasets; on the held-out vaccinia dataset its balanced accuracy was 0.58, close to chance. A housekeeping-gene panel (0.82) and random gene sets of the same size (0.74) did about as well as the translation-family genes (0.79), so this analysis does not show that the gene set is specific. Co-expression network edges reflect correlated expression across samples, not physical or regulatory interactions. This study is computational and hypothesis-generating; it does not claim primary discovery or causal function.

Conclusion

PoxHostAtlas is a reusable resource whose results can be independently reproduced. Across vaccinia infection and Myxoma M003 effector datasets, the host shows a reproducible program of inflammatory and NF-κB induction and cell-cycle suppression, with a directional decrease of ribosomal-protein genes; host shutoff is the clearest signal within vaccinia infection. Because this program recurs, at least directionally, in datasets from three independent studies and two host species, it is less likely to be a batch artifact, although other poxviruses, hosts and laboratories still need to be tested. We avoid family-first bias by analyzing the whole genome; within that program, DHX15 has the strongest pooled helicase evidence, whereas within the two vaccinia infection datasets DHX29 is the better supported of the two (meta-FDR 0.006 versus 0.075 for DHX15), so both are candidates for experimental testing. As more host-response datasets become available, the same openly released pipeline can incorporate them and refine these rankings.

Data and Code Availability

The entire analytical framework, from the analysis scripts and database, methods for data retrieval, differential analysis and meta-analysis results, and analyses (including figures), can be found in the project's repository along with the SHA-256 file for integrity authentication at https://github.com/mubarsad012/poxhostatlas. All input data are public. Count matrices for GSE278320 [1], GSE287860 and GSE288000 [9] were obtained from GEO; for GSE284044 [8] and GSE185520 [10], the authors' processed tables in GEO (DESeq2 and FPKM/$\log_2$FC tables, respectively) were used instead of raw counts. The single-cell metadata (HeLa and BJ5ta cells) and HeLa proteomics data used for external validation were taken from the supplementary tables of the Matía et al. preprint [7], not from GEO.

Funding

The authors received no specific funding for this work.

Conflicts of Interest

All authors declare no conflict of interest.

Ethics statement

This study is a reanalysis of publicly available, previously published data from experiments in established cell lines. No new experiments involving human participants, human tissue or animals were performed, and no ethical approval was required.

Author contributions

M.M. conceived and designed the study, identified and curated the datasets, wrote the analysis code, performed all analyses, produced the figures and tables, and wrote the manuscript. D.N. supervised the project, advised on analytical direction and on the interpretation of the results, and reviewed and edited the manuscript. Both authors read and approved the final manuscript.

Acknowledgments

We thank the OpenBio Student Research Institute for supporting this mentored research project, and the research groups who generated and publicly released the datasets reanalysed here. M.M. thanks D.N. for mentorship and guidance throughout the project.

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